

SKILL - 7

Process Synchronization and Child Process Monitoring Using waitpid().

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution.

```
dhars@DHARSHAN: /mnt/c/L x + v
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\dhars> wsl
dhars@DHARSHAN: /mnt/c/Users/dhars$ cd ossp
dhars@DHARSHAN: /mnt/c/Users/dhars/ossp$ nano skill7.c
dhars@DHARSHAN: /mnt/c/Users/dhars/ossp$ gcc skill7.c -o skill7
dhars@DHARSHAN: /mnt/c/Users/dhars/ossp$ ./skill7
===== SKILL 7: WAITPID AND PATH SEARCH =====

1. PATH Environment Variable

PATH = /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Eclipse Adoptium/jdk-2
1.0.12.101-hotspot/bin:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program F
iles (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v
1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs:/mnt/c/Users/dhars/AppData/Loca
l/Microsoft/WindowsApps:/mnt/c/Users/dhars/Downloads/jdk-25_windows-x64_bin:/mnt/c/Users/dhars/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/dha
rs/AppData/Local/Programs/Antigravity IDE/bin:/mnt/c/Program Files (x86)/MongoDB Atlas CLI:/mnt/c/Users/dhars/AppData/Roaming/npm:/snap/bin

2. Creating Child Process
Parent process started.
Parent PID: 852
Waiting for child using waitpid()...
Child process started.
Child PID: 853
Child process finished.

Child process has terminated.
Child exited normally.
Child exit status: 5

3. PATH Executable Search
```

```
Searching PATH for: ls
Executable found: /usr/bin/ls

Searching PATH for: gcc
Executable found: /usr/bin/gcc

Searching PATH for: notacommand
Command 'notacommand' not found in PATH.

===== SKILL 7 COMPLETED =====
dhars@DHARSHAN: /mnt/c/Users/dhars/ossp$ |
```